

SEMESTER END EXAMINATIONS – MARCH 2022

Program	: B.E. : Computer Science and Engineering	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: CS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit

UNIT- I

- Explain the various characteristics of an algorithm. CO1 (05)
 - Write an algorithm to add two polynomials stored in a single array. Demonstrate the working of the algorithm considering the following polynomials and store the sum in array A.
 $A(x) = 4x^{10} - 7x^6 - 2x^3 + 3x + 24$
 $B(x) = 12x^7 + 5x^6 + 7x^4 + 4x$ CO1 (10)
 - Write a recursive C function to print the elements of the array in reverse order. CO1 (05)
- Explain how sparse matrices can be stored efficiently in memory. Write a function to find the transpose of a sparse matrix stored in this way. CO1 (08)
 - Write an ADT for a string and include necessary string operations. CO1 (08)
 - Write a C macro to swap 2 integer numbers and explain. CO1 (04)

UNIT – II

- Why do we need dynamic arrays for stack implementation? Justify your answer with an example. CO2 (07)
 - Demonstrate the postfix expression evaluation using stack (in tabular form) with an example. CO2 (08)
 - Compare stack and queue operations with appropriate examples. CO2 (05)
- Compare and contrast Simple queue and Circular queue. CO2 (07)
 - Explain how infix expression is converted to postfix expression using stack? Illustrate with an example. CO2 (08)
 - Write C functions to implement multiple stacks push() and pop() operations. CO2 (05)

UNIT – III

- Write C functions to perform insertion and deletion of a node in singly linked list. CO3 (08)
 - Demonstrate the polynomial addition using linked lists with appropriate examples. CO3 (07)
 - Show the sparse matrix representation using linked list. CO3 (05)
- Write C functions to perform addition and deletion of elements in multiple queue using linked lists. CO3 (08)
 - Write a C function to invert a singly linked list. Show the steps through example. Also mention the diagram. CO3 (07)
 - Compare singly linked list, circular linked list and doubly linked list. CO3 (05)

UNIT – IV

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| 7. | a) | Construct the binary tree stepwise from its inorder (CBDEAFHGI) and postorder (CEDBHIGFA) traversals. | CO4 | (06) |
| | b) | Illustrate with an example, insertion of a node into a threaded binary tree. | CO4 | (08) |
| | c) | Illustrate insertion of an element into a min heap with a suitable example. | CO4 | (06) |
| 8. | a) | Construct a binary search tree stepwise for the following list of numbers: 24, 12, 6, 18, 30, 54, 7, 10, 42,36. Perform inorder, preorder and postorder traversal on the constructed binary search tree by writing the recursive functions for the same. | CO4 | (10) |
| | b) | Differentiate between max heap and min heap with an example. | CO4 | (04) |
| | c) | Explain the winner trees with a suitable example. | CO4 | (06) |

UNIT – V

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| 9. | a) | Explain the following with an example:
i) Height-biased Leftist Trees
ii) Weight-biased Leftist Trees. | CO5 | (08) |
| | b) | Compare adjacency list representation of graph representation with adjacency multilist representation with an example graph. | CO5 | (05) |
| | c) | Illustrate with an example, balancing of tree using four types of rotations, during insertion in AVL tree. | CO5 | (07) |
| 10. | a) | Write a C function of a biconnected component of a graph. | CO5 | (08) |
| | b) | Show the operations of single and double ended priority queues with appropriate examples. | CO5 | (07) |
| | c) | Write a C function of BFS of a graph. | CO5 | (05) |
